

ADAM CLAY

Curriculum Vitae

Office Address: University of Manitoba
Department of Mathematics
313 St. Paul's College
Winnipeg, R3T 2M6
Manitoba, Canada
Email Address: Adam.Clay@umanitoba.ca
Homepage: <https://adamjclay.github.io/>
Citizenship: Canadian
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Education

2010 Ph.D. University of British Columbia, Mathematics (advisor: Dale Rolfsen)
2005 M.Sc. University of British Columbia, Mathematics (advisor: Dale Rolfsen)
2003 B.Sc. University of King's College, Halifax, Canada, Mathematics

Employment

2024 – Full Professor, University of Manitoba
2018 – 2024 Associate Professor, University of Manitoba
2013 – 2018 Assistant Professor, University of Manitoba
2012 – 2013 CIRGET postdoctoral fellow, Université du Québec à Montréal (advisor: Steve Boyer)
2010 – 2012 NSERC postdoctoral fellow, Université du Québec à Montréal (advisor: Steve Boyer)

Administrative positions

Jan. 2026 – present Department Head, Dept. of Mathematics, University of Manitoba
Jan. 2026 – June 2026 Associate Head Graduate, Dept. of Mathematics, University of Manitoba
July 2025 – Dec. 2025 Acting Head, Dept. of Mathematics, University of Manitoba
Jan. 2020 – Jul. 2025 Associate Head Graduate, Dept. of Mathematics, University of Manitoba

Research interests

Group theory, low dimensional topology, algebraic properties of left- and circularly orderable groups, actions of groups on S^1 and $\mathbb{R}$, connections with dynamics, descriptive set theory, and 3-manifolds.

Publications

Books

1. *Ordered groups and Topology*, (with Dale Rolfsen) Graduate Studies in Mathematics 176 (2016), American Mathematical Society, Providence RI, USA.
2. *Circularly orderable groups*, (with Ty Ghaswala), in preparation.

Articles in progress

3. *A combinatorial approach to obstructing left-orderability of Dehn fillings*, (with Junyu Lu and Tjay Staruch), in preparation.
4. *The isomorphism problem for finitely generated bi-orderable groups*, (with Filippo Calderoni), submitted.
5. *Order-detection and non-left-orderable surgeries on links*, (with Junyu Lu), submitted.
6. *Generalized torsion in amalgams*, (with Tommy Cai), submitted.

Peer-reviewed journal articles

7. *Rigidity of braid group actions on $\mathbb{R}$ and of low-genus mapping class group actions on S^1* , (with Idrissa Ba and Tyrone Ghaswala), 33 pages. To appear in Annales de L'Institut Fourier.
8. *Order-detection, representation-detection, and applications to cable knots*, (with Junyu Lu), Transactions of the American Mathematical Society, 379 (3) (2026), 1755–1798.
9. *Ordered bases, invariant orderings and bi-orderable link groups*, (with Tommy Cai and Dale Rolfsen), Math. Proc. Camb. Phil. Soc., 180 (1) (2026), 219–235.
10. *Circular orderability of 3-manifold groups*, (with Idrissa Ba), Algebraic and Geometric Topology 25 (2025) 791–825.
11. *Graph manifolds that admit arbitrarily many Anosov flows*, (with Tali Pinsky), Mathematische Annalen 391 (2025), 5375–5390.
12. *The Borel complexity of the space of left-orderings, low-dimensional topology, and dynamics*, (with Filippo Calderoni), Journal of the London Mathematical Society 110 (2024), e70024.
13. *Condensation and left-orderable groups*, (with Filippo Calderoni), Proceedings of the American Mathematical Society Series B 11(50) (2024), 579–588.
14. *Cofinal elements and fractional Dehn twist coefficients*, (with Tyrone Ghaswala), International Mathematics Research Notices 9 (2024), 7401–7420.
15. *Order-detection of slopes on the boundaries of knot manifolds*, (with Steve Boyer), Groups, Geometry and Dynamics 18 (2024), no. 4, 1317–1348.
16. *The number of locally invariant orderings of a group*, (with Ian Thompson and Idrissa Ba), Journal of Group Theory 26 (2023), 1003–1021.
17. *A theorem of Mumford and Ramanujam for universal algebras*, (with R. Padmanabhan), Algebra Universalis 83 (2022), Article number: 38.
18. *Borel structures on the space of left-orderings*, (with Filippo Calderoni), Bulletin of the London Mathematical Society 54 (2022), 83–94.
19. *The space of circular orderings and semiconjugacy*, (with Idrissa Ba), Journal of Algebra, 586 (2021), 582–606.
20. *Circular ordering direct products and the obstruction to left-orderability*, (with Tyrone Ghaswala), Pacific Journal of Mathematics, 312 (2021), no. 2, 401–419.
21. *Promoting circular-orderability to left-orderability*, (with Jason Bell and Ty Ghaswala), Annales de L'Institut Fourier, 71 (2021), no. 1, 175–201.
22. *Generalizations of the Burns-Hale theorem*, Communications in Algebra, 48 (2020) no. 11, 4846–4858.
23. *Dense orderings in the space of left-orderings of a group*, (with Tessa Reimer) Canadian Mathematical Bulletin, 63 (2020), no. 2, 393–404.

24. *Free products of circularly ordered groups with amalgamated subgroup*, (with Ty Ghaswala) J. Lond. Math. Soc., 100 (2019), no. 3, 775–803.
25. *On the number of circular orders on a group*, (with Kathryn Mann and Cristobal Rivas) J. Algebra 504 (2018), 336–363.
26. *Automorphisms acting on the left-orderings of a bi-orderable group*, (with Sina Zabanfahm) New York J. Math. 23 (2017), 67–83.
27. *Foliations, orders, representations, L-spaces and graph manifolds*, (with Steve Boyer) Adv. Math. 310 (2017), 159–234.
28. *Testing bi-orderability of knot groups*, (with Colin Desmarais and Patrick Naylor) Canadian Mathematical Bulletin, 59 (2016), no. 3, 472–482.
29. *Graph manifolds, left-orderability and amalgamation*, (with Tye Lidman and Liam Watson) Algebr. Geom. Topol. 13 (2013), no. 4, 2347–2368.
30. *Left-orderability and exceptional Dehn surgery on two-bridge knots*, (with Masakazu Teragaito) Geometry and topology down under, 225–233, Contemp. Math., 597, Amer. Math. Soc., Providence, RI, 2013.
31. *Left-orderings and quotients of the braid groups*, J. Knot Theory Ramifications 21 (2012), no. 14, 125–130, 9 pp.
32. *Ordered groups, eigenvalues, knots, surgery and L-spaces*, (with Dale Rolfsen) Math. Proc. Cambridge Philos. Soc. 152 (2012), no. 1, 115–129.
33. *Left-orderable fundamental groups and Dehn surgery*, (with Liam Watson) Int. Math. Res. Not. IMRN 2013, no. 12, 2862–2890.
34. *Free lattice-ordered groups and the space of left-orderings*, Monatsh. Math. 167 (2012), no. 3–4, 417–430.
35. *On cabled knots, Dehn surgery, and left-orderable fundamental groups*, (with Liam Watson) Math. Res. Lett. 18 (2011), no. 6, 1085–1095.
36. *Exotic left orderings of the free groups from the Dehornoy ordering*, Bull. Aust. Math. Soc. 84 (2011), no. 1, 103–110.
37. *Isolated points in the space of left orderings of a group*, Groups Geom. Dyn. 4 (2010), no. 3, 517–532.
38. *A new characterization of Conrad’s property for group orderings, with applications*, (with Andrés Navas and Cristóbal Rivas, my contribution is the appendix) Algebr. Geom. Topol. 9 (2009) 2079–2100.
39. *Corrigendum to “On ordering free groups” [J. Symbolic Comput. 40 (2005) 1285–1290]*, (with Lawrence H. Smith) J. Symbolic Comput. 44 (2009), no. 10, 1529–1532.
40. *Densely ordered braid subgroups*, (with Dale Rolfsen) J. Knot Theory Ramifications 16 (2007), no. 7, 869–877.

Non-peer-reviewed work

41. *Left-orderability of 3-manifold fundamental groups*, CMS Notes 48 (2016) no. 2, 14–15.
42. *Orderable groups and bundles*, (with Mathieu Anel) (2012) 13 pages. arXiv:1208.5844
43. *Slope detection, foliations in graph manifolds, and L-spaces*, (with Steve Boyer) (2015), 30 pages. arXiv:1510.02378.

Lectures and presentations

Conferences

- 2026 June Saint John, Canada (CMS special session in geometric group theory): Borel complexity and isomorphism of bi-orderable groups
- 2025 Dec. Toronto, Canada (CMS special session in topology): Slope detection in knot complements and the L-space conjecture.
- June Quebec City, Canada (CMS special session in topology): Generalized torsion in 3-manifold groups.
- 2024 April Stillwater, USA (Redbud topology conference): Introduction to orderable groups and topology, Fractional Dehn twist coefficients and left-orderings of mapping class groups (two lectures).
- 2023 July Atlanta, USA (Tech topology summer school): Orderable groups and 3-manifolds (minicourse, five lectures).
- June Madrid, Spain (Orderings and Groups, my lecture was delivered online): Borel complexity and spaces of orderings.
- Apr. Montreal, Canada (Orderable groups): Introduction to order structures in groups (minicourse, four lectures).
- 2020 Oct. El Paso, USA (AMS special session in topology): The L-space conjecture, left-orderability and circular orderability of fundamental groups.
- 2019 Dec. Toronto, Canada (CMS special session in topology): Creating actions on the line from actions on the circle
- 2018 Dec. Vancouver, Canada (CMS special session in topology): Circular orderings and amalgamation
- 2017 Aug. Quito, Ecuador (XXII Coloquio Latinoamericano de Álgebra): Automorphisms acting on the space of orderings
- July Montreal, Canada (Mathematical Congress of the Americas): The number of circular orderings of a group
- 2016 Apr. Fargo, USA (AMS sectional meeting): Orderability and Dehn fillings of knot complements
- 2015 July Vancouver, Canada (PIMS Symposium on the Geometry and Topology of Manifolds): Foliations of graph manifolds
- Apr. Victoria, Canada (The 54th Cascade Topology Seminar): Foliations, left-orderability and L-spaces
- 2014 Sept. Santiago, Chile (Orderable Groups): Ordered groups and topology (minicourse, three lectures)
- Aug. Dublin, Ireland (The 10th William Rowan Hamilton Geometry and Topology Workshop): Graph manifolds, left-orderability and L-spaces
- June Winnipeg, Canada (CMS special session on low-dimensional topology and geometric group theory): Bi-orderability of fundamental groups of 3-manifolds
- 2013 Dec. Ottawa, Canada (CMS special session on geometric group theory and low-dimensional topology): The Burns-Hale theorem and its generalizations
- 2012 Dec. Montreal, Canada (CMS special session on geometrical group theory): Graph manifolds, orderability, L-spaces and foliations
- June Regina, Canada (CMS special session on geometry and topology): Foliations of graph manifolds and left-orderability
- Feb. Banff, Canada (BIRS workshop “Ordered groups and topology”): Left-orderability and foliations

- 2010 Apr. Banff, Canada (Cascade topology seminar): Group orderings and knots
 2009 Jan. Washington DC, USA (AMS special session on orderings and logic): Limit points in the space of left orderings of a group
 2008 May Montreal, Canada (Colloque Panquébécois des Étudiants): Cofinal elements in orderable groups
 2007 June Vancouver, Canada (Pure Math Graduate Student Conference): Densely ordered braid subgroups
 — May Banff, Canada (BIRS workshop): Normal subgroups of the braid groups and the Dehornoy order

Seminars

- 2026 June McMaster (Geometry and Topology Seminar): A combinatorial approach to left-orderability and Dehn filling
 2025 Mar. Prairie Mathematics Colloquium: Orderable groups and the L-space conjecture.
 — Feb. Dalhousie University (Colloquium): Orderable groups and the L-space conjecture.
 — Jan. University of Manitoba (Learning Seminar in Geometry and Topology): Farrell’s theorem using categories.
 2024 Feb. University of Manitoba (Learning Seminar in Geometry and Topology): Introduction to Thompson’s group F .
 2023 Oct. University of Winnipeg (Mathematics Seminar): Groups of homeomorphisms of $\mathbb{R}^n$.
 — Oct. University of Manitoba (Rings and Modules Seminar): Left-orderable groups and countable Borel equivalence relations.
 — Apr. Université du Québec à Montréal (CIRGET Seminar): Rigidity of braid group actions on the real line.
 — Mar. University of Manitoba (Rings and Modules Seminar): Analogs of Mumford-Ramanujam Theorem for Universal Algebras (Part II) Structure Theorems.
 2021 Oct. Oklahoma State University (Topology Seminar): Circular orderings, 3-manifolds and covering spaces.
 2020 Oct. The University of British Columbia (Topology Seminar): Cohomological conditions for left-orderability.
 — June Group actions seminar (Online): Towards a cohomological characterization of left-orderability
 2019 Nov. University of Chicago at Illinois (Logic seminar): The structure of the space of left-orderings
 — Oct. University of Manitoba (Learning seminar in geometry and topology): Groups of homeomorphisms of low-dimensional manifolds
 — May Technion (Geometry and topology seminar): Promoting group actions on the circle to actions on the real line
 — May University of British Columbia (Algebra and topology seminar): When does a group that acts on the circle also act on the line?
 2018 Dec. Université du Québec à Montréal (CIRGET working group): Introduction to circular orderings
 — Nov. University of Texas at Austin (Geometry and topology seminar): Circular orderings, amalgamation and 3-manifolds
 2017 Nov. University of Manitoba (Learning seminar in geometry and topology): Free subgroups of groups of homeomorphisms
 — Sept. University of Manitoba (Learning seminar in geometry and topology): The Ping-Pong Lemma

- Aug. Dalhousie University (Colloquium): An introduction to circular orderings and left orderings of groups
- Mar. The University of British Columbia (Topology seminar): The spaces of left and circular orderings of a group
- 2016 July Dalhousie University (Colloquium): Smooth Foliations of 3-Manifolds
- 2015 Nov. University of Manitoba (Learning seminar in geometry and topology): Introduction to the space of left-orderings (2 talks)
- Sept. University of Manitoba (Learning seminar in geometry and topology): Introduction to group growth
- Mar. University of British Columbia (Graduate class on orderable groups): Foliations and left-orderings of Seifert fibred manifolds (2 talks)
- 2014 Oct. University of Manitoba (Learning seminar in geometry and topology): Dehn surgery and fundamental groups of manifolds (2 talks)
- Mar. University of Manitoba (Colloquium): Knot groups and orderability
- Feb. University of Manitoba (Learning seminar in geometry and topology): Introduction to braids and knot theory (3 talks)
- 2013 May Dalhousie University (Colloquium): Ordered groups and 3-manifolds
- Mar. Université du Québec à Montréal (CIRGET geometry and topology seminar): Orderability and knots
- Feb. University of Manitoba (Colloquium): Ordered groups and 3-manifolds
- 2012 Oct. McMaster University (Geometry and topology seminar): Left-orderability, foliations, L-spaces and graph manifolds
- 2011 July University of British Columbia (Algebra and topology seminar): Left-orderability and Dehn surgery
- 2010 Nov. Université du Québec à Montréal (CIRGET geometry and topology seminar): Orderability and knots
- 2009 Nov. University of Tokyo (Topology seminar): An introduction to orderable groups
- Jan. Dalhousie University (Graduate student seminar): An introduction to orderable groups
- 2008 Nov. Universidad de Santiago de Chile (Dynamical systems seminar): Isolated points in the space of left orderings of a group
- Feb. University of British Columbia (Algebra and topology): The space of left orders of the braid groups
- 2007 Nov. Laboratoire de Mathématiques Nicolas Oresme (Special seminar): The space of orderings and the braids
- Nov. University of British Columbia (Algebra and topology seminar): Normal subgroups of the braid groups and the Dehornoy order

Organizing conferences, meetings and activities

- 2026 Co-organizer, CMS session in geometric group theory, June 5–8 Saint John, Canada.
- 2025 Co-organizer, CMS session in low dimensional topology, December 5–8 Toronto, Canada.
- 2023 Co-organizer, CRM conference on orderable groups, April 24–May 5, Montreal, Canada.
- 2021 Co-organizer, CMS session in group theory and low dimensional topology, December 3–6 Vancouver, Canada.
- Co-organizer, CMS session in low dimensional topology, June 5–8 Ottawa, Canada.

- Co-chair of organizing committee, Nexus 2021: Bridging Academia and Industry, May 19–21, Winnipeg, Canada.
- 2019 Co-organizer, 63rd Cascade Topology seminar, September 14–15 Winnipeg, Manitoba.
- Co-organizer, Ordered Groups and Rigidity in Dynamics and Topology (BIRS/CMO event 19w5044), June 16–21 Oaxaca, Mexico.
- 2017 Co-organizer, CMS session in low dimensional topology and geometric group theory, December 8–11 Waterloo, Canada.
- 2016 Co-organizer, CMS session in low dimensional topology and geometric group theory, December 3–5 Niagara Falls, Canada.
- 2015 Co-organizer, CMS session in low dimensional topology and geometric group theory, December 6–9 Montreal, Canada.
- 2014 Co-organizer, CMS session in low dimensional topology and geometric group theory, June 6–9 Winnipeg, Canada.
- 2013 Co-organizer, CMS session in geometric group theory and low dimensional topology, December 6–9 Ottawa, Canada.
- Co-organizer, Low dimensional topology, knots and orderable groups, July 1–5 Luminy, Marseille.

Editorial activities

- 2025 – Associate Editor, Canadian Mathematical Society Journals (CMB, CJM, CMC)

Referee and review activities

Journals refereed (many multiple times)

- Canadian Mathematical Bulletin
- Canadian Journal of Mathematics
- Topology and its Applications
- Algebra Universalis
- Order – A Journal on the Theory of Ordered Sets and its Applications
- Proceedings of Winterbraids
- Compositio Mathematica
- Journal of Knot theory and its Ramifications
- Communications in Analysis and Geometry
- Journal of Algebra
- Algebraic & Geometric Topology
- Communications in Algebra
- Proceedings of the American Mathematical Society
- Journal of the London Mathematical Society
- Proceedings of the London Mathematical Society
- Publicationes Mathematicae Debrecen
- Proceedings of the Edinburgh Mathematical Society
- Commentarii Mathematici Helvetici
- Journal of Topology
- Pacific Journal of Mathematics
- Journal of Group Theory
- New York Journal of Mathematics

Granting agencies refereed

Fonds de recherche du Québec - Nature et technologies (FRQNT)
 Natural Sciences and Engineering Research Council (NSERC)
 Swiss National Science Foundation (SNSF)

Academic honours and grants

2025 Fellow of the Canadian Mathematical Society
 — CRM-Simons Professor (July-August)
 2023 University of Manitoba URGP (\$6000)
 — CRM-Simons Scholar (April-May)
 — PI for PIMS conference grant (\$2500)
 2020 – 2025 NSERC Discovery grant (\$243000 over 9 years)
 2019 PI for PIMS conference grant (\$4500)
 2018 – 2019 University of Manitoba URGP (\$7500)
 2014 – 2015 University of Manitoba URGP (\$6000)
 2014 – 2020 NSERC Discovery grant (\$114000 over 5 years)
 2013 – 2016 University of Manitoba startup grant (\$45000)
 2013 Co-PI for NSF conference grant 1305741 (\$32000 USD)
 — PI for PIMS conference grant (\$4000)
 2010 – 2012 NSERC postdoctoral fellowship, Université du Québec à Montréal
 2008 University graduate fellowship, University of British Columbia
 2005 – 2007 NSERC PGS-D, University of British Columbia
 2003 – 2005 NSERC PGS-M, University of British Columbia

Awards

2021 University of Manitoba Merit Award in Service
 2017 Faculty of Science Award for Innovation in Teaching
 2008 University of British Columbia graduate teaching award (departmental)
 2003 King's Medal, University of King's College (institutional award)

Media

Linking algebra and topology (The Manitoban research interview, 2025)
 Math Talk! (YouTube interview series, 2024)

Teaching activities

2026 Winter Algebra 3 (Math 3322 at the University of Manitoba)
 2025 Fall Introduction to Topology (Math 3390 at the University of Manitoba)
 — Orderable groups (Reading course at the University of Manitoba)
 2025 Summer Geometric Group Theory (Reading course at the University of Manitoba)
 2024 Fall Introduction to Topology (Math 3390 at the University of Manitoba)
 — Introduction to Descriptive Set Theory (Reading course at the University of Manitoba)
 2024 Summer Geometric Group Theory (Reading course at the University of Manitoba)
 2024 Winter Introduction to Differential Geometry (MATH 7360 at the University of Manitoba)
 2023 Summer Languages, Automata and Groups (Reading course at the University of Manitoba)

- 2022 Fall Geometric Group Theory (Reading course at the University of Manitoba)
 — Introduction to Topology (Math 3390 at the University of Manitoba)
- 2022 Winter Orderable groups and topology (Reading course at the University of Manitoba)
- 2021 Fall Introduction to Topology (Math 3390 at the University of Manitoba)
 — Homology and Cohomology (Reading course at the University of Manitoba)
- 2021 Summer Bass-Serre Theory (Reading course at the University of Manitoba)
- 2021 Winter Advanced Galois Theory (Reading course at the University of Manitoba)
- 2020 Fall Introduction to Topology (Math 3390 at the University of Manitoba)
- 2020 Winter Introduction to Encryption (SCI 2000 at the University of Manitoba)
- 2019 Fall Introduction to Analysis (MATH 2080 at the University of Manitoba)
- 2018 Winter Algebra 3 (MATH 3322 at the University of Manitoba)
 MATH 4190 Topological groups in Physics (Reading course at University of Manitoba)
- 2017 Fall Introduction to Analysis (MATH 2080 at the University of Manitoba)
 Differential Calculus (Math 1500 at the University of Manitoba)
- 2017 Winter Analysis 3 (MATH 3472 at the University of Manitoba)
 Differential Calculus course coordinator (Math 1500 at the University of Manitoba)
- 2016 Fall Topics in Geometric group theory (Reading course at the University of Manitoba)
 Honours Differential Calculus (Math 1230 at the University of Manitoba)
- 2016 Winter Algebra 1 (MATH 2020 at the University of Manitoba)
 Differential Calculus (Math 1500 at the University of Manitoba)
- 2015 Fall Introduction to Topology (Math 3390 at the University of Manitoba)
- 2015 Winter Engineering Analysis 1 (Math 2132 at the University of Manitoba)
 Advanced Calculus (Math 3760 at the University of Manitoba)
- 2014 Fall Advanced Calculus (Math 3760 at the University of Manitoba)
- 2014 Winter Topology I (Math 3240 at the University of Manitoba)
 Differential Calculus (Math 1500 at the University of Manitoba)
- 2012 Fall Enriched Linear Algebra and Geometry (Math 134 at McGill)
- 2011 Fall Linear Algebra and Geometry (Math 133 at McGill)
- 2010 Fall Linear Algebra and Geometry (Math 133 at McGill)
- 2009–2010 Organizer of nine calculus workshops (Math 110 at UBC)
- 2005–2008 Ordinary differential equations (Math 265 at UBC)
- 2005 Winter Integral calculus (Math 103 at UBC)

Postdoctoral

- Idrissa Ba (2019–2022)
 Ty Ghaswala (2017–2019)
 Juliana Theodoro de Lima (2015–2016, co-supervisor with Steve Boyer)

PhD

- Junyu Lu (2021–2025)
 Thesis: Order Detection and left-orderability of 3-manifold groups
- Tommy Cai (2020–2025)
 Thesis: Orderability and generalized torsion.

MSc

Tjay Staruch (Jan 2025–present)
 Eliza Haraldson (Sept 2024–present)
 Graeme Henrickson (Jan 2024–present)

Dana Kapoostinsky (2022–2024)
 Thesis: Regular orderings of left orderable groups.

Vindya Lakshani (2021–2022)
 Withdrew from program due to covid travel restrictions.

Irshad Ayoob (2017–2020)
 Thesis: Ordered groups and a theorem of Farrell.

Adriana Ciupeanu (2016–2019)
 Thesis: Left-orderable groups and amenability.

Serhii Dovhyi (2014–2017)
 Thesis: Non-left-orderable surgeries of twisted torus knots.

Ali Khardani (2013–2014, joint supervision with Steve Boyer at UQAM)
 Thesis: Groupes ordonnés et les variétés de dimension 3.

Undergraduate students mentored

2026 USRA	Nauman Khan	<i>The number of circular orderings of a group</i>
2023 Indep.	Eliza Haraldson	<i>Left-orderability and smooth CBERS</i>
2022 Indep.	Eliza Haraldson	<i>Left-orderability and countable Borel equivalence relations</i>
2022 USRA	Dana Kapoostinsky	<i>Left-orderability and homeomorphisms of the disk</i>
2021 USRA	Waweru Kariuki	<i>Bass-Serre theory and circular orderings</i>
2021 USRA	Tjay Staruch	<i>Left-orderability and fundamental groups of cyclic branched covers of knots</i>
2021 USRA	Dana Kapoostinsky	<i>Obstructing circular orderability</i>
2019 USRA	Robert Harris	<i>Circular orderability and obstructing left-orderability</i>
2019 USRA	Mitchell Heron	<i>Left-orderability and Dehn filling</i>
2019 USRA	Mary-Elaine Herrera	<i>Left-orderings and branched cyclic covers of knots</i>
2019 USRA	Ian Thompson	<i>Locally invariant orderings and amenability</i>
2018 USRA	Jeremy Friesen	<i>Groups and topological dynamics</i>
2018 USRA	Robert Harris	<i>Density of representations of the free group</i>
2018 USRA	Tessa Reimer	<i>Isolated points in spaces of orderings</i>
2018 Hons.	Jeremy Friesen (honours project)	<i>Topological groups in physics</i>
2017 USRA	Tessa Reimer	<i>Algebraic properties of ordered groups</i>
2016 USRA	Sina Zabanfahm	<i>Spaces of left-orderings of braid groups</i>
2015 USRA	Patrick Naylor	<i>Diffuse groups and locally invariant orderings</i>
2015 USRA	Colin Desmarais	<i>A brute force approach to knot group bi-orderability</i>
2014 USRA	Patrick Naylor	<i>Bi-orderability of two bridge knot groups</i>
2014 USRA	Colin Desmarais	<i>Testing bi-orderability of knot groups</i>

Seminars organized or co-organized

2023–2025	Learning seminar in geometry and topology (at University of Manitoba, co-organizer)
2017–2019	Learning seminar in geometry and topology (at University of Manitoba, co-organizer)

- 2014–2016 Learning seminar in geometry and topology (at University of Manitoba, co-organizer)
- 2013 Co-organizer, CIRGET-LaCIM seminar (at UQÀM)
- 2011 Working group in low dimensional topology (at UQÀM)

Committee work

Professional committees

- NSERC Evaluation group 1508 (Group chair, 2025-present)
- Canadian Mathematical Society Invested Funds Committee (member 2023–present)
- Canadian Mathematical Society Finance Committee (member 2016–2021, chair 2022–present)
- NSERC Evaluation Group (member 2020, Pure Mathematics co-chair and chair 2021–2022)
- NSERC Discovery Horizons Program Fit Advisor (2023)
- Canadian Mathematical Society board of directors (2020–2023)

Administrative committees

- University of Manitoba Senate (2022–present)
- University of Manitoba Tri-Agency bridge funding committee (2022–2025)
- Faculty of Science awards selection committee (2024)
- Hiring (U of M departmental committee) (2024)
- Chair of mathematics instructor hiring committee (June 2024)
- Research committee (U of M departmental committee) (2021–2023)
- Faculty of Science Nucleus Tenure Committee (U of M faculty committee) (2020–2022)
- Hiring (for Faculty of Science administrative position) (2022)
- Hiring (U of M departmental committee) (2021)
- Data Science M.Sc. program committee (U of M faculty committee) (2020–2022)
- Graduate studies (U of M departmental committee) (2018–2020)
- Promotion and tenure (U of M departmental committee) (2018)
- Hiring (U of M departmental committee) (2017)
- Promotion (U of M departmental committee) (2016)
- Executive (U of M departmental committee) (2016–2019)
- Scholarships and awards (U of M departmental committee) (2016–2019)
- Hiring (U of M departmental committee) (2015)
- Research advisory team science (U of M faculty committee) (2014–2017)
- Hiring (U of M departmental committee) (2014)
- Strategic planning (U of M departmental committee) (2013–2016)
- Various comprehensive examination committees (15 committees over 2013–2026)
- Chair of various examination committees (10 committees between 2013–2026)